



Effect of alloying elements on microstructure and properties of multiprincipal elements high-entropy alloys

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ABSTRACT

Ten high-entropy alloys are prepared by an arc furnace. The microstructure and the properties of the alloys are investigated. FeNiCrCuCo and FeNiCrCuMo alloys consist of a single FCC solid solution. When Cu or Co in the alloys is substituted by Al, the microstructures of the alloys change to that of BCC solid solution or that of a BCC + FCC solid solution. The results show that Cu with a FCC structure promotes the formation of a FCC solid solution. Although Al has a FCC structure, it does not help the formation of a FCC solid solution. In addition, when Zr is added into the alloys, complicated compounds are present due to the stronger compound formation tendency between Zr and other components. It is found that the hardness of alloys with a BCC structure is higher than that of the alloys with a FCC structure. The hardness of the alloys increases with increasing of Al content. When complicated compounds are formed, the hardness increases further. The highest hardness reaches 566 HV due to the strengthening of the second phase precipitation.

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1. Introduction

It is known that when the principal element has a high solubility, the strength of the solution can increase due to the existence of the solution hardening. If a solution structure is kept, the alloy would have a better combination of strength and plastics. As the amount of alloying element increases, intermetallic compounds however are typically formed in the alloys due to the limited solid solubility of the alloys, which leads to enhanced strength at the expense of accompanying embrittlement. To realize the goal of a solution structure, a new alloy design concept, multi-principle-element alloy or high-entropy (HE) alloy, was explored by Yeh et al. based on the thermodynamic principle, which consists of a simple FCC or BCC structure [1–3]. HE alloys composed of normally n major alloying elements with $n \geq 5$ in equimolar or near-equimolar ratios have better stability of the solution due to the contribution of the mixing entropy. Following Boltzmann's hypothesis on the relationship between entropy complexity [2], the configurational entropy change, ΔS_{conf} , during the formation of a solid solution from n elements with equimolar fractions, may be calculated from

the following equation:

$$\Delta S_{\text{conf}} = -R \ln \left(\frac{1}{n} \right) = R \ln(n) \quad (1)$$

where R is the ideal gas constant. When $n = 5$, $\Delta S_{\text{conf}} = 1.61R$, which approaches the size of the melting entropy of most intermetallic compounds (about $R-2R$). As results, intermetallic compounds are absent and a single solid solution is built with structures of a single FCC, a single BCC or FCC + BCC [2–3]. Due to the above structural and compositional characteristics, the HE alloys are ductile and working-hardenable, and have high strength up to 800 °C [3–5]. Thus, the alloys have potential applications as high-temperature structure materials and work tools. HE alloys were mostly obtained by rapid solidification, mechanical alloying, CVD and PVD methods respectively. The properties such as strength, hardness, wear resistance and corrosion resistance of the alloys were investigated [6–11]. In the HE alloys, the effects of adding elements content (such as Al, B, N, Cr, Ti, Fe and V) on the structure and properties have been considered [12–18]. The alloying elements with different lattice structures will affect the microstructures and the properties of the alloys. So far, it is not clarified however whether there is a relationship between the structure of the alloying elements and that of the HE alloys. This question is especially true when Al as an alloying element is added into the HE alloys. This is partly induced by the dualism of metal and nonmetal characteristics of Al due to its special electronic structure. Al has a nonmetal characteristic when

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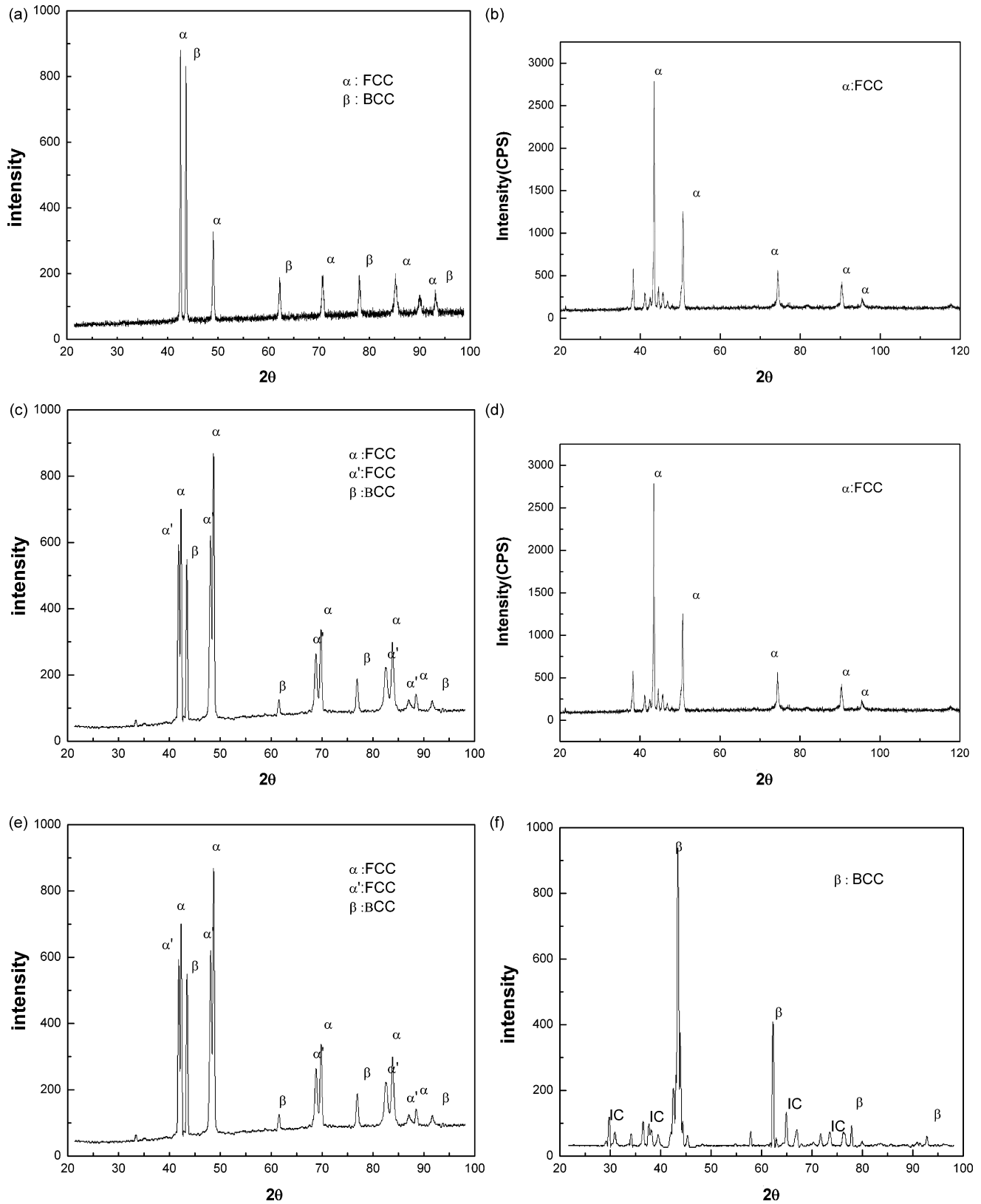


Fig. 1. X-ray diffraction analysis of the alloys. (a) FeNiCrCuCo, (b) FeNiCrCuAl, (c) FeNiCrCuAl, (d) FeNiCrCuMo, (e) FeNiCrCuMn, and (f) FeNiCrCuZr.

Table 1
Basic crystal structural properties of selected metallic elements at room temperature [19–21]

Element	Cu	Al	Co	Ni	Cr	Fe	Mn	Mo	Zr
Crystal structure	FCC	FCC	HCP	FCC	BCC	BCC	BCC	BCC	HCP
Atomic radius (Å)	1.413	1.582	1.385	1.377	1.423	1.411	1.428	1.550	1.771
Lattice constant (Å), <i>a</i> (<i>b</i>)	2.556	2.863	2.407 (2.507)	2.492	2.498	2.482	2.24	2.725	3.179 (3.231)

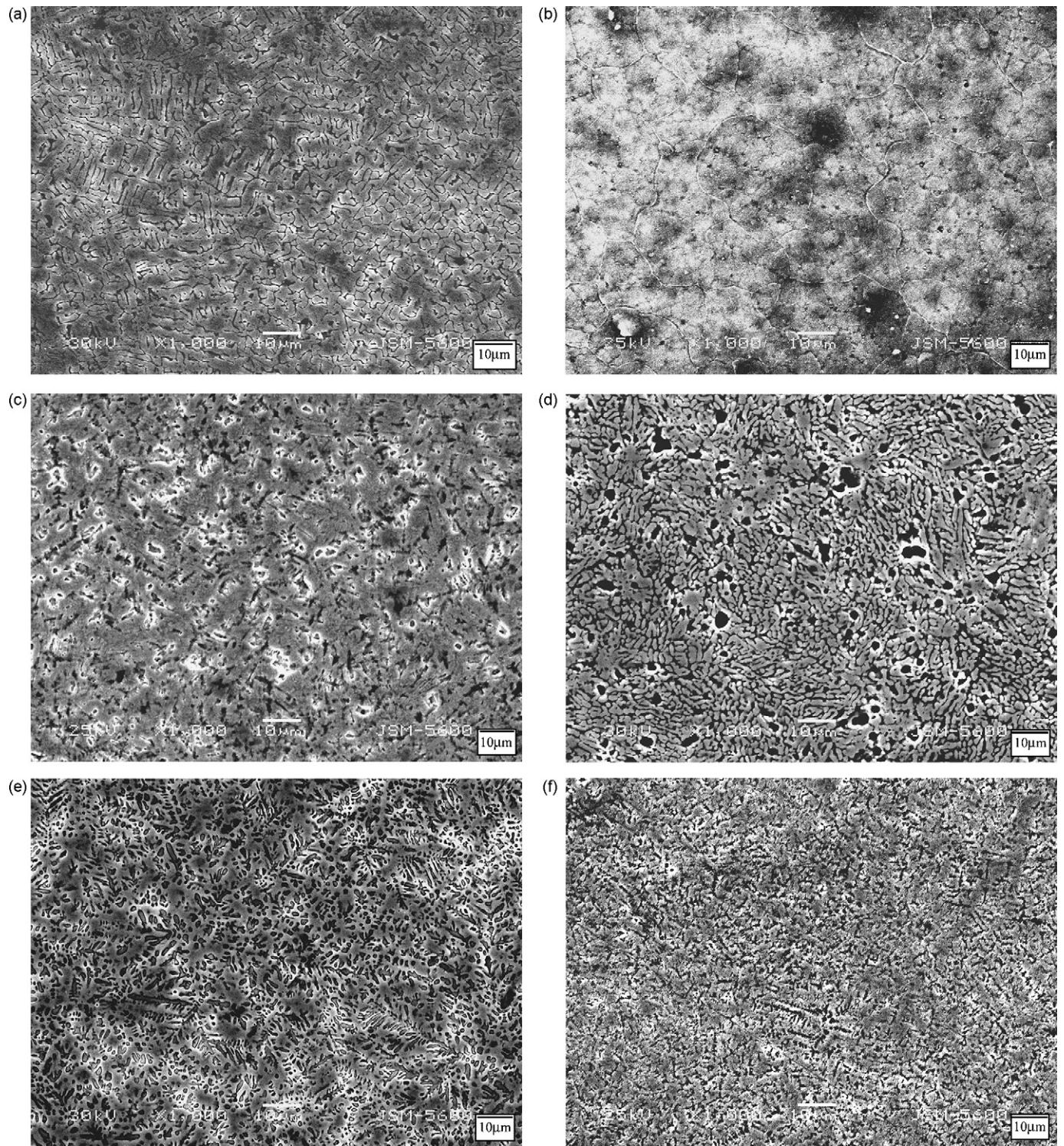


Fig. 2. Optical and SEM micrographs of the alloys. (a) FeNiCrCuCo, (b) FeNiCrCoAl, (c) FeNiCrCuAl, (d) FeNiCrCuMo, (e) FeNiCrCuMn, and (f) FeNiCrCuZr.

Table 2

The composition of the alloys used in the experiment (at%)

Alloy	Fe	Ni	Cr	Cu	Co	Mo	Mn	Zr	Al
FeNiCrCuCo	20	20	20	20	20	–	–	–	–
FeNiCrCuAl	20	20	20	20	–	–	–	–	20
FeNiCrCuMo	20	20	20	20	–	20	–	–	–
FeNiCrCuMn	20	20	20	20	–	–	20	–	–
FeNiCrCuZr	20	20	20	20	–	–	–	20	–
FeNiCrCoAl	20	20	20	–	20	–	–	–	20
FeNiCrCoAl _{1.5}	18.2	18.2	18.2	–	18.2	–	–	–	27.2
FeNiCrCoAl ₂	16.7	16.7	16.7	–	16.7	–	–	–	33.2
FeNiCrCoAl _{2.5}	15.4	15.4	15.4	–	15.4	–	–	–	38.4
FeNiCrCoAl ₃	14.3	14.3	14.3	–	14.3	–	–	–	42.8

Al is added into metals while Al has a metal characteristic if Al is alloyed with nonmetal elements.

In this contribution, the alloys with different alloying elements, especially Al, are fabricated and their microstructures and mechanical properties are measured. Through these studies, we wish to know the effect of alloying elements on the structures and properties of the HE alloys.

2. Experimental methods

The rod and slice-shaping ingot of the alloy are obtained in a cold copper hearth with a cooling rate of about 10^4 K/s. The diameter of rod is 6 mm while the slice thickness is 1 mm. The rod samples are used as observed microstructure and measured properties. Table 1 lists structural constants of the related alloying elements while Table 2 shows the theoretical compositions of the alloys. The composition design of the alloys is made as follows: Among five principle elements, Fe, Ni and Cr as essential elements are always remained. Other two elements can be changed by case to identify their effect on crystalline structures of HE alloys.

The phase structure analysis of the samples was made by Rigaku D/max 2500 X-ray diffractometer at 50 kV and 250 mA. Scanning angles ranged from 25° to 135° with a scanning rate of $5^\circ/\text{min}$.

The microstructure of the alloys was observed using optical microscope (OP) and a scanning electron microscope (SEM).

The hardness of the samples was determined by a Vickers hardness instrument. Vickers' load was 200 g with a loading time of 15 s. The density was determined by use of Archimedes's principle by a balance with a precision of 0.01 mg.

3. Results and discussion

X-ray phase analysis results for the fabricated alloys with different kinds of elements are shown in Fig. 1. The corresponding microstructures are shown in Fig. 2. When a HE alloy consists of

elements near IIIV family group in the Periodic Table, such as FeNiCrCuCo and FeNiCrCuMo alloys, a single FCC structure is present, as shown in Fig. 1a and d. When Al is present, which substitutes Cu, such as FeNiCrAlCo alloy shown in Fig. 1b and a BCC structure arises. However, when this substitution of Al is changed from Cu to Co, namely FeNiCrCuAl alloy, a mixed BCC + FCC structure is present. The microstructure of FeNiCrCuMn alloy that Mn element substitutes for Co element in FeNiCrCuCo alloy is FCC + BCC structure. The microstructure of FeNiCrCuZr alloy that Zr element add to the alloy is BCC solid solution and intermetallic compounds. The results imply that Cu with a FCC structure promotes the formation of a FCC solid solution. Although Al has a FCC structure, it does not help the formation of a FCC solid solution. In addition, when Zr is added into the alloys, complicated compounds are present due to the stronger chemical combination tendency between Zr and other components. Fig. 2 presents the microstructure of the alloys with different elements. The results show that the microstructure is a typical dendritic crystal structure. The double phases of the complex structure exists in dendritic and interdendritic site respectively, their respective distribution need to be further distinguish by TEM in next work. The microstructure of the alloys with Al content becomes coarse.

In order to investigate effect of Al on the microstructure and the properties of the alloys, the alloys with different Al contents are fabricated. The X-ray phase analysis results of these alloys with different Al content are shown in Fig. 3. All alloys consist of BCC phase with a super structure peak as shown in early literatures [17,22]. The intensity of the diffraction peak decreases with increasing of Al content because the addition of Al induces aberrance of the crystal lattice. BCC phase is consisted of a structure ($A_2 + B_2$) during cooling, the similar result was reported by Ref. [7]. We consider that Al and Ni as essential components due to the high formation enthalpy H^c should hold the sites of center and corner in B_2 , respectively. While the other components substitute Al and Ni depending on their relative differences of the electronegativity value I and the atomic radius r . Due to the similar I and r values of Co, Ni and Fe in 1.88, 1.91, 1.83, and in 1.25 Å, 1.25 Å, 1.24 Å, respectively, Fe and Co could take up the site of Ni in B_2 structure. On the other hand, I and r values of Cr and Al are 1.61, 1.66, and 1.43 Å, 1.25 Å, and Cr may occupy the site of Al. The lattice constant of the alloys increases from 2.878 to 2.887 nm with increasing of Al content since the atomic radius of Al is larger than other elements in the alloys as shown.

Fig. 4 presents the microstructure of the alloys with different Al contents. The results show that the microstructure is a typical dendritic crystal structure. The microstructure of the alloys becomes coarse with increasing of the aluminum content. The increase of Al content not only introduces the lattice strain, but also affects the microstructure and the properties of the alloys. When Al molar ratio increases from 1 to 3, the microstructure modality varies from a shape of cotton-boll, or that of a petal to a fishbone shape respectively. Also, the grains coarsen with the increasing of the Al content.

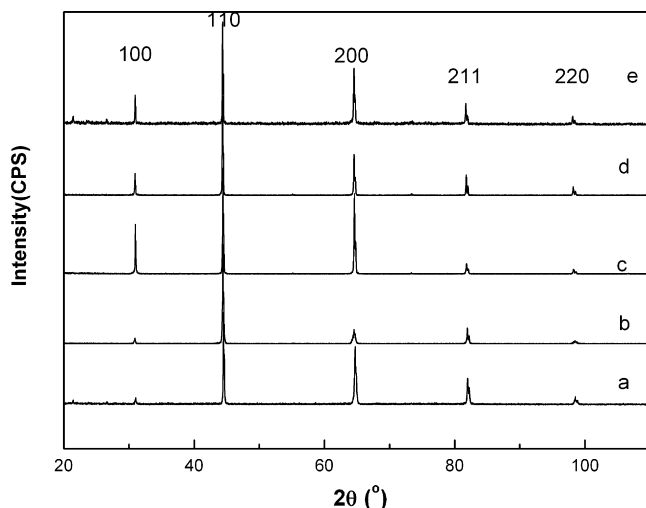


Fig. 3. X-ray diffraction analysis of the alloys with different Al at%. (a) FeNiCrCoAl alloy, (b) FeNiCrCoAl_{1.5} alloy, (c) FeNiCrCoAl₂ alloy, (d) FeNiCrCoAl_{2.5} alloy, and (e) FeNiCrCoAl₃ alloy.

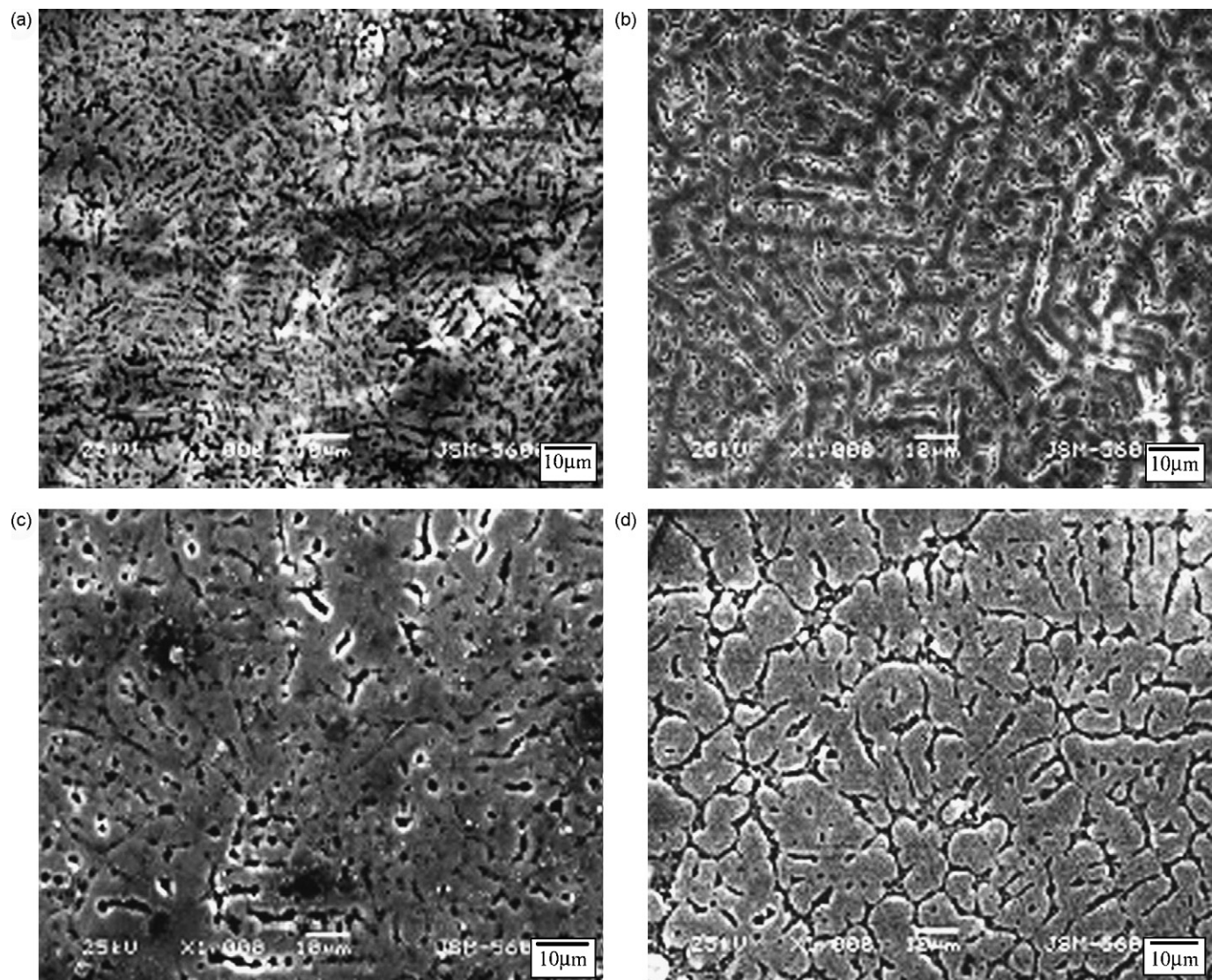


Fig. 4. SEM micrographs of the alloys with different Al at%. (a) FeNiCrCoAl_{1.5} alloy, (b) FeNiCrCoAl₂ alloy, (c) FeNiCrCoAl_{2.5} alloy, and (d) FeNiCrCoAl₃ alloy.

Table 3 shows the hardness of alloys. The result shows that the alloys with BCC structure possess higher hardness than that of the alloys with FCC structure. The hardness of the alloys with the compounds is highest and reaches 566 HV due to the strengthening of the second phase precipitation, but the compounds present in the alloys adding Zr content and lead to brittleness increase. The hardness of the alloys increases with Al content increasing. It is known that the chemical mixing enthalpy of Al–Ni, Al–Co,

Al–Fe and Al–Cr are -30 kJ mol^{-1} , -24 kJ mol^{-1} , -10 kJ mol^{-1} and -6 kJ mol^{-1} respectively, however the chemical mixing enthalpy of Cr–Ni, Cr–Co, Cr–Fe, Co–Fe, Ni–Fe and Co–Ni are 3 kJ mol^{-1} , 8 kJ mol^{-1} , 13 kJ mol^{-1} , 13 kJ mol^{-1} , 12 kJ mol^{-1} and 17 kJ mol^{-1} respectively [23]. This result implies that the combinative force of Al element with other elements in the alloy is strongest among the combinative force of the elements in the alloys. In addition, atomic size of Al is biggest in the all elements of the alloys, so that the aberrance of crystal lattice increases with Al content increasing. As the Al with larger atomic radius occupies the lattice sites, the lattice distortion energy will increase significantly and the effect of solid solution strengthening is enhanced, thus the strength greatly increases.

4. Conclusions

FeNiCrCuCo and FeNiCrCuMo alloys consist of a single FCC solid solution. When Cu or Co in the alloys is substituted by Al, BCC or BCC + FCC structures are present. Cu or Al, which have a FCC structure, promotes or deteriorates the formation of a FCC solid solution. In addition, when Zr is added into the alloys, complicated compounds are present due to its strong compound formation tendency

Table 3
The hardness of the alloys

Alloy	Structure	Hardness (HV)
FeNiCrCuCo	FCC	286
FeNiCrCuMo	FCC	263
FeNiCrCuAl	FCC + BCC	342
FeNiCrCuMn	FCC + BCC	296
FeNiCrCoAl	BCC	395
FeNiCrCoAl _{1.5}	BCC	402
FeNiCrCoAl ₂	BCC	432
FeNiCrCoAl _{2.5}	BCC	487
FeNiCrCoAl ₃	BCC	506
FeNiCrCuZr	BCC + compounds	566

with other components. The hardness of alloys with a BCC structure possesses is higher than that of the alloys with FCC structure. When complicated compounds form, both hardness and brittleness increase further due to the strengthening mechanism of the second phase precipitation. The hardness of the alloys increases with increasing of Al content.

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